Harsh Chhallani

EXTC

Company: Interactive Brokers

Job Profile: Junior Software Engineer

Offer Type: Placement Only

CTC: 14.65 + Performance Bonus (Cash+/Stock component) (15-30 %)

Location: Mumbai

Placement Process:

1. **<u>Online Assessment</u>** [Students with >=7 CGPA]

The first stage was an **online assessment** consisting of a single DSA question + 4 logic/Aptitude questions. The DSA question was mainly based on mathematical reasoning rather than complex data structures.

After this round, **24 students were shortlisted**.

2. **<u>Predictive Index Assessment</u>** [24 students]

The shortlisted candidates were asked to complete a **Predictive Index assessment** the following day. This assessment was focused more on behavioural and personality evaluation rather than technical skills.

After this stage, **13 candidates were shortlisted for the interview process**.

3. **<u>Technical Interview Round 1 (Online)</u>** [13 students]

The first technical round was conducted online on a coding platform called **Meetl**.

The interview focused on:

- **Core Java fundamentals**
- **Java Collections Framework**
- **Object-Oriented Programming concepts**
- **DBMS basics**
- **Data Structures and Algorithms**

I was given **two DSA problems**:

- One was Leetcode easy level.
- The second problem was of **medium difficulty (roughly LeetCode medium level)**.

I had to **write the code for both problems and explain the logic clearly**.
After this round, **3 out of the 13 candidates were shortlisted** for the next round.

<u>4. Technical Interview Round 2 (Onsite) [3 students]</u>

This round was conducted **onsite at the company office** and was a **pen-and-paper coding round**.

This interview again focused heavily on **OOP concepts and Java fundamentals**.

Some of the questions included:

- Explanation of Parent p = new Child() and Child c = new Parent() along with reasoning.
- Typecasting in objects.
- Lambda functions in Java.
- Streams in Java.
- Serializable interface.
- Questions about my **hackathon experiences and projects**.

After this , the interviewer asked **4** DSA questions

1. Linked List

Delete a node in a singly linked list in **constant time**.

2. Array Problem

Find the **second maximum element** in an array using **one pass and constant space**.

3. Integer Stream Problem

The interviewer described a scenario where **a large stream of integers flows into a queue throughout the day**, with the constraint that the **difference between consecutive numbers is always 1**.

At the end of the day, we need to answer q **queries asking whether a specific number x appeared in the stream**, and the requirement was to answer queries in **constant time and constant space**.

4. Array Partition

Given an array, divide the elements into **two arrays such that the difference between their sums is minimum** . (time complexity of order 1)

The problems themselves were not difficult but were **slightly tricky and required reasoning/proof .**

I wrote code for all four problems and explained the logic clearly. The interviewer seemed satisfied with both my **OOP explanations and DSA solutions**.

The next day I received a call from HR saying that **the feedback was positive**, and I was invited for the next round.

5. Technical Interview Round 3 (Senior Engineering Manager) [2 students]

This round was again conducted **onsite and a pen-and-paper** round .

The interviewer was an **Engineering Manager who had been with the company for around 11 years**.

The round began with questions on the **Java Collections Framework**, after which the discussion went deep into **HashMap internals**.

I had to explain:

- How key-value pairs are stored
- Load factor
- Rehashing
- Collision handling
- How collision handling worked **before Java 8 and after Java 8**

After this discussion, I was asked to write a **Student class** with attributes:

- name
- id
- marks

I wrote the class and implemented the constructor.

Then the interviewer asked whether we could **directly use this class as a key in a HashMap**.

This question was meant to check my understanding of **equals() and hashCode()**.

I explained:

- The default implementations provided by the **Object class**
- How they compare memory references
- Why overriding them is necessary for logical equality

I then implemented:

- equals() using **student id**
- hashCode() accordingly

I also explained what would happen if we **did not override these methods**.

Next we discussed:

- Comparable vs Comparator

- Writing a Comparable implementation
- Writing a Comparator for a different custom sorting strategy
- What the return values **-1, 0, and 1** represent

The interviewer then moved to **JDBC**, which I had not worked with before, so I honestly mentioned that I had not used JDBC.

We then discussed **SQL concepts**, including:

- Different types of joins
- Database indexing
- Data structures used in indexing (B-Trees)

Finally, the interviewer asked **two logical puzzles**:

Puzzle 1: 25 Horses and 5 Tracks

Find the **top three fastest horses**.

Puzzle 2: 2 Eggs and 100 Floors

Determine the highest floor from which an egg can be dropped without breaking.

The interviewer was mainly evaluating **how I approached the problems and broke them down logically**, rather than expecting an immediate answer.

I struggled a bit with the **horses puzzle**, but with some hints I eventually arrived at the solution.

I had mixed feelings after this round because the puzzle part did not go perfectly.

6. **<u>Technical Interview Round 4 (Online) [2 students]</u>**

After about a week, HR informed us that there would be **one final technical round**.

This round was conducted **virtually** and had **two interviewers**:

- One senior engineering leader from the **US**
- One engineer from the **Mumbai office (if selected , he/she will be your manager)**

This round was more **conversational compared to previous rounds**.

We started with a discussion about my **projects and hackathon experiences**. I was asked about the most challenging aspects of my projects and how I approached solving them.

One interesting question was about **a product in the market that I admire or would like to work on**, and I spoke about a technology product I had seen on Shark Tank India.

The discussion then moved into **database concepts**, including:

- Relational vs Non-relational databases
- MongoDB
- Redis (Since I had used in my project)
- Why Redis, despite being very fast, is usually not used as a primary database for large systems

I was also given a **probability puzzle** involving **50 black balls and 50 white balls distributed across two bags**, where the goal was to maximize the probability of drawing a black ball.

I initially took some time to reason about the problem, and with a hint from the interviewer I arrived at the correct solution. Towards the end of the interview, the discussion became more open-ended. I asked about a concept the interviewer mentioned called the **S2 geometry data structure**, which is used for geospatial indexing in systems like Google Maps.

I also asked what areas a new engineer should focus on during the **first few months after joining**.

Overall I was able to answer all questions in this round , but again felt dumb because I took a lot more time than required on that simple probability puzzle , brain felt saturated at that moment and stopped thinking, but with one hint from interviewer , arrived at the ans .

After this round, there was a long silence from the company for around **14–17 days**. During this time, **another candidate** who was interviewing along with me received a call from HR for the HR discussion round**, while I did not receive any update.**

Naturally, I felt that my process might have ended there. However, after around two weeks, I unexpectedly received a call from HR informing me that **my profile had been moved to a different team** internally, and because of that, **I would have to go through two additional technical &/ team rounds** which were scheduled immediately on the **next day** .

7. **Technical Interview Round 5 (Online)**

The fifth round was conducted by a manager from the Mumbai office with around 7 years of experience. The interview was relatively short and focused mainly on **Java fundamentals** and **problem solving**.

He first asked me about the internal working of **HashMap**, where I explained **buckets**, **key-value storage**, **hashCode()**, **equals()** , **collisions**, **collision handling**, etc.

Then he moved to a DSA problem similar to the "Search Suggestions System" problem on LeetCode.

Similar to : [Leetcode Search Suggestions]

I came up with a sorting + string prefix matching based solution. O(n^2) and coded the approach and output was correct.

Although the problem had a more optimized binary search based approach possible, he did not ask me to optimize further once the logic and output were correct . The round ended after he asked whether I had any questions for him.

8. **<u>Technical Interview Round 6 (Online)</u>**

The sixth and final technical round was with a senior engineer from the US team who had around 15 years of experience. This round was very different from the previous ones.

It was not focused on coding or rapid-fire technical questions. Instead, it was more of a deep two-way discussion about the team, their **domains**, **systems**, and **the type of work they do**. He spent a significant amount of time explaining the team structure and the backend systems they work on.

I asked a few thoughtful questions regarding the domain and one specific question that he appreciated a lot during the discussion. I also asked him what I should focus on learning before joining so that I could reduce the amount of training effort required from the team side once I joined. He seemed to like that question as well, and the overall interaction was very positive and engaging. The conversation lasted for around one hour.

After these rounds, I received a call from HR stating that all interviewer feedback was **positive** and that **the final HR discussion round** was scheduled.

The HR round mainly covered resume overview, family background, relocation, future plans, joining timeline, and compensation discussion.

Toward the end of the HR round, I was also introduced to the manager(same person who took my second last interview) of the team I would potentially join, and we discussed the team goals and what I should focus on before joining.

After the HR round, there was another waiting period because the profile had to go through final approval from the US side. Around a month later, I finally received the official confirmation and offer from the company for the role of Junior Software Engineer.

Useful Tips:

Maintain cgpa above 7.5 or 7 at least, I wasn't able to sit in many good companies for this reason .

Simple to-dos :

1) 2-3 good projects and in depth understanding of these projects .
2) DSA with any lang , but know the fundamentals and core of the lang well .
3) OOPS along with implementation. (You will not forget and understand better when you code) .
4) SQL 50
5) DBMS

OS , CCN are asked less comparatively , do them after above topics .

Best of luck .

Feel free to connect for any qs .

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